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General Ozochiawaeze

Education

Purdue University
Ph.D. Candidate in Mathematics 2021 – 2027 (expected)
- Research Interests: Inverse Problems & Partial Differential Equations
- Advisors: Peijun Li & Isaac Harris

New Jersey Institute of Technology
M.S. in Applied Mathematics May 2021
- Advisor: Christina Frederick

Rutgers University
B.A. in Mathematics May 2018

Publications & Preprints

1. R. Ayala, I. Harris, **G. Ozochiawaeze.** Factorization method for the biharmonic scattering problem for an absorbing penetrable obstacle. *Submitted,* 2025. arXiv:2511.05711
2. I. Harris, P. Li, **G. Ozochiawaeze.** Sampling methods for the inverse cavity scattering problem of biharmonic waves. *Under Review, Inverse Problems*, 2025. arXiv:2509.02773

Work Experience

MIT Lincoln Laboratory | Lexington, MA May 2023 – Oct 2023
- *Summer Research Intern, Group 36 – Integrated Missile Defense Technology*
- Applied compressive sensing to enhance radar imaging for missile target detection, improving reconstruction accuracy.
- Presented compelling research findings on compressed radar sensing, demonstrating potential applications in enhanced defense technologies.

MIT Lincoln Laboratory | Lexington, MA May 2022 – Aug 2022
- *Summer Research Intern, Group 37 – Advanced Undersea Systems & Technology*
- Developed algorithms to analyze coherence loss in sonar signals, improving underwater detection and tracking.
- Applied sonar array interferometry to extract spatial information from acoustic wave-fronts.

Teaching

- Lecturer, **Purdue University —** *MA 16020: Applied Calculus II*, Fall 2025
- Recitation Instructor, **Purdue University —** *MA 261: Multivariable Calculus*, Spring 2024
- Teaching Assistant, **Purdue University —** *MA 511: Linear Algebra and Applications*, Spring 2023
- Recitation Instructor, **Purdue University —** *MA 261: Multivariable Calculus*, Fall 2022
- Recitation Instructor, **Purdue University —** *MA 162: Calculus II*, Spring 2022
- Recitation Instructor, **Purdue University —** *MA 161: Calculus I*, Fall 2021
- Instructor, **Accel Learning —** Mentored students in preparation for AMC and AIME math competitions, Spring–Fall 2020
- Math Instructor, **Mathnasium**, K-12 Teaching & Tutoring, Fall 2018–Spring 2019

Awards and Honors

- **SIAM (Society of Industrial & Applied Mathematics) Student Travel Award** 2025
- **NSF Computational Mathematics Program Award**, DMS-2208256, 2023-2024
- **GEM Fellowship** 2022
- **Graduate C/Startup Research Grant** 2020
- **Tau Sigma Honors Society** 2016

Skills

Computing & Software: `Python`, `C++`, `MATLAB`, `COMSOL`, `FEniCS`

Seminar & Conference Talks

A Factorization Method Approach to the Biharmonic Transmission Problem in Absorbing Media (invited speaker) 2025
SIAM Conference on Analysis of Partial Differential Equations (PD25), Pittsburgh

Localizing Cavities in Biharmonic Scattering with Limited Data 2025
CCAM Lunch Seminar, Purdue University

Extending Qualitative Reconstruction to Biharmonic Scattering with Limited Data 2025
Summer School: Theory and Applications of Elliptic PDE, UC Irvine

Extending Qualitative Reconstruction in Biharmonic Scattering with Limited Data 2025
Graduate Student Analysis Seminar, Purdue University

On Sampling Methods for Recovering a Clamped Cavity in a Thin Plate 2025
CCAM Lunch Seminar, Purdue University

A Linear Sampling Method for Recovering a Clamped Cavity in a Thin Plate 2024
Purdue Graduate Research Day, Purdue University

A Linear Sampling Method for Recovering a Clamped Cavity in a Thin Plate 2024
SIAM Student Chapter Conference, Purdue University

Comparing and Integrating the Probe Method and Method of Singular Sources 2023
Graduate Student Analysis Seminar, Purdue University

The Factorization Method in Inverse Scattering 2023
Graduate Student Analysis Seminar, Purdue University

Compressive Radar Imaging in Missile Defense 2023
Integrated Missile Defense Technology Group, MIT Lincoln Laboratory

Quantifying Real Operational Performances of SONAR Arrays in Random Ocean environments 2022
Advanced Undersea Systems & Technology Group, MIT Lincoln Laboratory

Finite Element Modeling of Underwater Acoustic Environments 2022
Purdue Math Student Colloquium, Purdue University

Conferences Attended

Summer School: Theory and Applications of Elliptic PDE 2025
UC Irvine

Mathematics and Statistics in Industry Workshop and Panel Discussion 2025
Purdue University

Conference on Mathematics of Wave Phenomena 2025
Karlsruhe Institute of Technology (KIT)

Paris-Saclay Conference in Analysis and PDE 2024
Laboratoire de Mathématiques d'Orsay

SIAM Student Chapter Conference 2024
Purdue University

Workshop on Scientific Computing 2022
Purdue University

New Ideas in Computational Inverse Problems 2022
BIRS, Banff International Research Station

Informal Talks

The Dawn of the Diophantine Era: Algebra at the Crossroads of Empires 2025
Math History Seminar, Purdue University

Fixed Points in Action: from Brouwer to Lefschetz & Beyond 2025
Topolodays, Purdue University

The Life, Mathematics, and Political Advocacy of Sofya Kovalevskaya 2024
Math History Seminar, Purdue University

The K-theoretic Atiyah-Singer Index Theorem 2024
Student Operator Algebras Seminar, Purdue University

Introducing Pseudodifferential Operators 2024
Purdue Math Student Colloquium

Bourbaki and the Foundations of Modern Math 2023
Math History Seminar, Purdue University

Wasan: Mathematical Treasures in Early Modern Japan 2022
Math History Seminar, Purdue University

Service

Purdue Math Department Graduate Representative 2025-2026
Purdue University

Mentor, Directed Reading Program in Computational Topology 2025
Purdue University — mentee: Aaron (Sang Hyun) Kim

Founder and Organizer, Math History Seminar 2022
Purdue University

Rutgers Undergraduate Mathematics Association Treasurer & Web Manager 2017-2018
Rutgers University